

WEATHER: RMSE at each forecast step
Lookback = 24 steps, Forecast horizon = 72 steps

Forecast step	CNN	LSTM	dCeNN+ELM	dCeNN+ELM+ASP
1.000	29.699	60.788	27.913	26.211
2.000	31.340	42.641	31.271	29.785
3.000	32.677	36.396	33.549	32.113
4.000	34.602	35.472	35.088	33.691
5.000	35.584	35.414	36.090	34.779
6.000	36.195	35.350	36.865	35.618
7.000	36.360	35.449	37.374	36.145
8.000	36.535	35.766	37.693	36.455
9.000	36.921	36.152	38.033	36.826
10.000	37.305	36.491	38.404	37.229
11.000	37.537	36.730	38.683	37.529
12.000	37.587	36.976	38.911	37.722
13.000	37.528	37.186	39.338	38.075
14.000	37.810	37.394	39.691	38.403
15.000	38.056	37.554	39.777	38.517
16.000	38.201	37.681	39.851	38.619
17.000	38.050	37.764	39.808	38.576
18.000	37.837	37.784	39.610	38.347
19.000	38.171	37.781	39.427	38.195
20.000	38.395	37.776	39.542	38.338
21.000	38.561	37.763	39.759	38.596
22.000	38.574	37.753	39.887	38.788
23.000	38.162	37.762	39.887	38.820
24.000	38.146	37.800	39.608	38.546
25.000	38.134	37.935	39.604	38.502
26.000	38.076	38.155	39.880	38.731
27.000	38.139	38.395	40.246	39.057
28.000	38.566	38.606	40.520	39.347
29.000	38.986	38.743	40.711	39.615
30.000	38.937	38.804	40.735	39.704
31.000	38.450	38.791	40.764	39.724
32.000	38.255	38.743	40.829	39.756
33.000	38.772	38.667	41.074	39.995
34.000	39.425	38.591	41.281	40.197
35.000	39.239	38.523	41.472	40.347
36.000	39.051	38.448	41.726	40.546
37.000	38.894	38.423	42.202	40.954
38.000	38.954	38.468	42.681	41.380
39.000	39.543	38.576	42.833	41.520
40.000	39.741	38.762	42.875	41.575
41.000	39.786	39.006	42.782	41.469
42.000	39.723	39.229	42.453	41.171
43.000	39.960	39.411	42.116	40.880
44.000	40.177	39.561	41.894	40.683
45.000	40.270	39.670	41.924	40.763
46.000	40.044	39.763	41.934	40.838
47.000	40.125	39.864	41.921	40.863
48.000	40.029	40.009	42.138	41.057
49.000	40.189	40.188	42.433	41.323
50.000	40.385	40.332	42.589	41.483
51.000	40.377	40.443	42.665	41.539
52.000	40.839	40.538	42.737	41.592
53.000	41.313	40.612	42.760	41.625
54.000	41.261	40.653	42.774	41.666
55.000	41.343	40.661	42.785	41.705
56.000	40.779	40.613	42.838	41.767
57.000	40.955	40.484	42.872	41.816
58.000	41.427	40.319	42.839	41.818
59.000	41.039	40.155	42.986	41.959
60.000	40.868	40.024	43.163	42.073
61.000	40.480	39.971	43.324	42.168
62.000	40.526	39.992	43.334	42.137
63.000	41.166	40.096	43.096	41.862
64.000	41.359	40.271	42.800	41.587
65.000	41.098	40.445	42.738	41.571
66.000	41.045	40.596	42.892	41.767
67.000	41.032	40.686	43.221	42.074
68.000	41.317	40.804	43.467	42.248
69.000	41.430	40.943	43.483	42.274
70.000	41.037	41.154	43.239	42.133
71.000	40.794	41.451	42.928	41.913
72.000	40.811	41.808	42.803	41.801

• This table shows how prediction error changes from the first forecast step to the last forecast step.
• It is especially useful for explaining whether a model degrades smoothly or sharply as it predicts further into the future.
• Lower is better. Large mistakes are penalized more strongly than in MAE.
• Weather combines multiple physical variables. Aggregate RMSE or MAE should therefore be read mainly as a model-comparison tool, not as a single physical-unit error.